

MECC Inspector Home Page (PCF)

Technical Setup & Troubleshooting Guide

Establishing a new environment and a new inspector user

Item	Value
Components covered	Online control: Home.MECCHomePage (v0.0.51) Offline control: MyNamespace.MyControl (v0.0.39)
Platform	Dynamics 365 Field Service / Power Apps Component Framework (PCF), React 17
Audience	CRM administrators, functional consultants, support (Level 1 / Level 2)
Purpose	Checklists for configuring a new environment and a new user, plus a symptom-to-cause troubleshooting matrix
Document version	1.0 — 2026-08-22

1. Overview

The Home Page control replaces the standard Dynamics home screen for inspectors. It shows the inspector's daily counters, a facility search, patrol start/end buttons, and the Multi-Type Inspection wizard that creates the Account, the Work Order and all of its related records.

Almost every problem reported as "the control does not work" is in fact missing configuration data, not a code defect. The control silently degrades: if a lookup or a configuration record is missing, the related button, list, or counter simply shows nothing. This document lists exactly which record and which field drives each element on the screen.

1.1 The two versions

	Online version (MECCHomePage)	Offline version (MyControl)
Namespace / constructor	Home.MECCHomePage	MyNamespace.MyControl
Version in manifest	0.0.51	0.0.39
Intended client	Browser / model-driven app (always connected)	Field Service Mobile with mobile offline enabled
Data access style	Uses \$expand and server-side query operators; relies on server plug-ins to create the child records of the Work Order	No \$expand, no link-entity, no 'ne' operator; each lookup is read with a separate request; child records are created from the client
Work Order children	Created server-side by plug-ins (Work Order Incident, Service Tasks, Questions, Penalties, Process Extension)	Created by the control itself (WOSTHelpers, ProcessExtensionHelpers) and flagged duc_iscreatedoffline = true so the plug-ins skip them after sync
Extra feature	Barcode scanning enabled; Project (type 16) search via Logic App (ProjectHelpers, endpoint not configured yet)	Pending Work Order Actions counter; patrol handling; offline-safe search
Work Order form opened after create	b7b3d199-8809-f111-8341-6045bd8e2841	b5ea80ff-301f-f111-88b1-6045bd8e2841

Important: both versions use the same configuration tables and the same user fields. Everything in sections 2 to 5 applies to both, unless the row is marked "offline only".

2. Tables the control reads and writes

If a security role does not grant the privilege below, the corresponding feature fails silently (empty list / zero counter) or ends with the generic message "Please contact your administrator".

Table (logical name)	Used for	Privilege needed
systemuser	Organizational unit, department, bookable resource, Arabic name, judicial authority flag	Read

Table (logical name)	Used for	Privilege needed
msdyn_organizationalunit	Unknown account, site account, default incident type, English name, hide-campaign flag	Read
bookableresource / bookingstatus	Auto-booking after Work Order creation	Read + Create (bookableresourcebooking)
bookableresourcebooking	"Scheduled inspections today" counter, booking list buttons, auto-booking	Read, Create
msdyn_workorder	Counters, Work Order creation, sub-status update	Read, Create, Write
msdyn_workorderincident	Primary incident line of the Work Order (offline only)	Create
msdyn_workorderservicetask	Service tasks / inspection form (offline only)	Create
msdyn_incidenttype	Incident type list, default Work Order type, process definition, department	Read
msdyn_incidenttypeservicetask	Service task templates (offline only)	Read
msdyn_inspection / msdyn_inspectiondefinition	Inspection template and its published version (offline only)	Read
duc_msdyn_incidenttype_msdyn_organizational	N:N incident type ↔ organizational unit mapping	Read
new_inspectioncampaign	Campaign list, patrol start/end, parent campaign	Read, Write
duc_organizationunitaccounttypes	Which inspection types appear in the wizard for the unit	Read
duc_accounttype	Numeric inspection type code and account-name override	Read
account	Facility search, search-or-create of the inspected subject	Read, Create, Write
duc_addressinformation	GPS coordinates captured at inspection start	Read, Create
new_license	Advanced search by licence number	Read
duc_inspectionaction	"Pending Work Order actions" counter (offline version)	Read
duc_processdefinition / duc_processextension / duc_processstage / duc_processstatus / duc_processsubstatus	Process extension created after the Work Order (offline only)	Read + Create (extension)

Table (logical name)	Used for	Privilege needed
duc_inspectionquestionconfiguration / duc_inspectionquestioncategories / duc_questionanswersconfiguration	Inspection questions copied to the service task (offline only)	Read + Create (answers)
duc_penalty / duc_workorderpenalties	Penalties copied to the service task (offline only)	Read + Create

3. Establishing a new environment

3.1 Deployment steps

1. Import the solution that contains all tables above and the two PCF controls.
2. Add the control to the required form / custom page: Home.MECCHomePage for the browser app, MyNamespace.MyControl for the offline mobile app. The control is bound to any single-line text field (sampleProperty) — the field value itself is not used.
3. Publish all customizations.
4. Update the environment-specific GUIDs listed in section 3.2 — this is mandatory, they are hard-coded in the source and will not exist in a new environment.
5. Rebuild and re-import the control after changing the GUIDs (npm run build, then push the solution).
6. Create the configuration data of sections 4 and 5.
7. Create the first user as described in section 6 and run the verification checks in section 8.

3.2 Hard-coded GUIDs that must be re-pointed

These values were captured from the source environment. In a new environment they will not resolve; the result is an empty counter, a button that opens an error, or a blank form.

Where	GUID	What it points to	Effect if wrong
Main.ts — counters	f16d80d1-fd07-4237-8b69-187a11eb75f9	Booking status "Scheduled"	"Scheduled inspections today" always 0
Main.ts (online only)	a2ad7f6a-f763-461a-b724-6d4371506baa	Booking status "In Progress"	In-progress bookings not counted
Main.ts — button	4073baca-cc5f-e611-8109-000d3a146973	Bookable resource booking view (today)	Button opens an error / empty list
Main.ts — button	f52b7a9b-cae8-f011-8406-6045bd9c20cb	Bookable resource booking view (all)	Button opens an error / empty list
Main.ts — button	bee0efc7-40e4-f011-8406-6045bd9c224c (offline) 960bbe64-7c09-f111-8341-6045bd8e23b3 (online)	Pending work orders view	Button opens an error / empty list
Main.ts — button	aad92cc9-00c6-f011-8544-000d3a2274a5	Completed work orders view	Button opens an error / empty list
MultiTypeInspection.ts	b5ea80ff-301f-f111-88b1-6045bd8e2841 (offline) b7b3d199-8809-f111-8341-6045bd8e2841 (online)	Work Order main form opened after creation	Work Order is created but the form does not open
ProjectHelpers.ts (online)	PROJECT_SEARCH_API_URL = ""	Logic App used by Project inspection (type 16)	Returns dummy sample projects until the URL is set

3.3 Option-set values the code depends on

Field	Value	Meaning used by the control
new_inspectioncampaign.duc_campaignstatus	1 / 100000004 / 2	Planned or Ended (patrol can be started) / In progress (patrol can be ended)
new_inspectioncampaign.duc_campaigninternaltype	100000004	Patrol campaign
new_inspectioncampaign.duc_campaigntype	100000000 / 100000004	Inspection campaign (shown in the wizard) / Patrol
msdyn_workorder.msdyn_systemstatus	690970001	Status set on every Work Order created by the control
msdyn_workorder.duc_taxtype	100000000 / 100000001	Income / Excise — filters which penalties are copied
duc_inspectionaction.duc_status	100000003, 100000005	Excluded from the "pending actions" counter
bookableresource.resourcetype	3	Resource of type User — required for auto-booking
msdyn_inspectiondefinition.msdyn_state	1	Published inspection template version
account.duc_accountinspectiontype / duc_accounttype.duc_accounttype	1 ... 16	Inspection type code — see section 5.2

4. Organizational unit configuration

The organizational unit of the user is the master key of the whole home page: inspection types, incident types, campaigns and patrols are all filtered by it. If the user has no organizational unit, the wizard opens but every dropdown is empty.

Field on msdyn_organizationalunit	Required?	Drives
duc_englishname	Yes	Unit name shown on the page; also used for the Natural Reserve / Wildlife behaviour (exact text match — see note below)
duc_unknownaccount	For anonymous inspections	Account used for inspection type 4 (Anonymous) and for the "Anonymous" checkbox on types 3, 6, 7. Empty ⇒ "Please contact your administrator"
duc_siteaccount	For site inspections	Account used for inspection type 12 (Site). Empty ⇒ "Please contact your administrator"
duc_defaultincidenttype	Recommended	Fallback incident type when the user selects neither an incident type nor a campaign
duc_hidecampaignonhomepage	Optional	When true, the Campaign field is hidden in the wizard for users whose duc_department points to this unit

Name matching note: the code compares duc_englishname with the literal strings "Inspection Section – Natural Reserves" (with an en dash –, not a hyphen), "Wildlife - Plant Life Section" and "Wildlife - Natural Resources Section". A different spelling, a trailing space or a hyphen instead of the dash disables the related behaviour.

4.1 Inspection types shown in the wizard

The list of inspection types is not an option set — it is built from the junction table duc_organizationunitaccounttypes. For every type the unit is allowed to inspect, create one row:

- duc_organizationunit = the organizational unit
- duc_accounttype = the account type record
- duc_name / duc_namear = the English / Arabic label shown in the dropdown

The numeric code comes from the related record: duc_accounttype.duc_accounttype. If that field is empty, the control falls back to -1 and the type behaves incorrectly (no identifier field, no account search). This is one of the most common configuration mistakes.

4.2 Inspection type codes and their required input

Code	Type	Field the inspector must fill	Account matching field
1	Vehicle	Plate number (+ colour, brand optional)	duc_accountidentifier
2, 15	Individual	Qatari ID (exactly 11 digits) + Name	duc_accountidentifier
3	Cabin	Qatari ID + Name (or Anonymous checkbox)	duc_accountidentifier

Code	Type	Field the inspector must fill	Account matching field
4	Anonymous	None	Unit's duc_unknownaccount
5, 7	Company / Manor	CR number or CP number	duc_accountidentifier / duc_cpnumber
6	Wilderness camp	Qatari ID + Name (or Anonymous checkbox)	duc_accountidentifier
10, 11	Establishment / Hospital	Registration number	duc_moinumber
12	Site	None	Unit's duc_siteaccount
13	Important maritime areas	None (always creates a new account)	—
14	Boat	Boat number	duc_accountidentifier
16	Project	Request / permit number (+ location details)	duc_accountidentifier

5. Incident type and process configuration

The incident type is the record that decides what the Work Order becomes. It must be configured completely, otherwise the Work Order is created but stays empty (no type, no service task, no questions, no penalties).

Field on msdyn_incidenttype	Purpose	If missing
duc_organizationalunitid (or an N:N row in duc_msdyn_incidenttype_msdyn_organizational)	Makes the incident type visible to the unit's users	The incident type never appears in the dropdown
duc_judicialauthorityisrequired	Hidden from users whose systemuser.duc_hasjudicialauth is not true	Type visible to everybody (when false or empty)
msdyn_defaultworkordertype	Sets msdyn_workordertype on the new Work Order	Work Order created with no Work Order Type
duc_processdefinition	Source of the process extension defaults (offline version)	No process extension is created; work order sub-status stays empty
Related msdyn_incidenttypeservicetask rows	Templates for the service tasks / inspection form	No service task, therefore no inspection form on the Work Order
msdyn_inspection on the service task template	The inspection template; the control picks its latest published version (msdyn_state = 1)	Service task created without an inspection form

5.1 Process definition (duc_processdefinition)

Used by the offline version to reproduce, on the device, what the server plug-ins normally do. The following lookups are written unconditionally — if any of them is empty, the process extension creation fails and the Work Order keeps an empty sub-status:

- duc_startstage → current stage of the process extension
- duc_defaultstatus → status
- duc_defaultsstatus → sub-status; its duc_value field must contain the GUID of a valid msdyn_workordersubstatus record in this environment
- duc_startaction, duc_targetentitycustomerlookupname, duc_targetentitysubjectname, duc_parentlookup → used for the target mapping

5.2 Questions and penalties (offline version)

- Questions: duc_inspectionquestionconfiguration rows whose duc_primaryincidenttype is the selected incident type. Each row must have duc_questionname and duc_questioncategory — rows without a category are skipped.
- Penalties: duc_penalty rows whose duc_incidenttype is the selected incident type. If the Work Order has duc_taxtype = Income (100000000) or Excise (100000001), only penalties with the same tax type are copied. duc_maximumamount is copied into duc_penaltyamount.
- Both are skipped if records already exist for the service task, so a repeated sync does not duplicate them.

6. Establishing a new user — checklist

Follow the steps in order. Steps 1 to 4 are mandatory for the home page to work at all.

#	Step	Where	Verification
1	Assign the security role that grants the privileges of section 2	System user	User can open the account and work order tables
2	Set duc_organizationalunitid (organizational unit)	System user	Unit name and inspection types appear on the home page
3	Create a bookable resource for the user: userid = the user, resourcetype = 3 (User)	Bookable resource	Auto-booking works after creating a Work Order
4	Link the resource back on the user: duc_bookableresourceid	System user	The three daily counters stop showing 0
5	Set duc_usernamearabic	System user	The welcome banner shows the Arabic name instead of "Inspector"
6	Set duc_department (if campaign visibility is controlled per department)	System user	Campaign field shown or hidden as expected
7	Set duc_hasjudicialauth = Yes for inspectors with judicial authority	System user	Incident types marked duc_judicialauthorityisrequired appear
8	Make sure a booking status named exactly "Scheduled" exists	Booking status	Auto-booking succeeds
9	Offline only: add the user to the Mobile Offline Profile and let the first sync finish	Mobile offline profile	The wizard works with the device in airplane mode
10	Assign at least one patrol campaign if the unit uses patrols (see section 6.1)	Inspection campaign	Start Patrol button appears

6.1 Conditions for the patrol button to appear

All conditions must be true on the same new_inspectioncampaign record:

- owninguser = the signed-in user (the campaign must be owned by the inspector, not by a team)
- duc_organizationalunitid = the user's organizational unit
- duc_campaigninternaltype = 100000004 (Patrol)
- duc_fromdate ≤ today ≤ duc_todate
- duc_campaignstatus = 1 or 100000004 → the button shows "Start Patrol"; = 2 → the button shows "End Patrol"

Note: patrol start/end is blocked while the device is offline; the control shows the offline message instead.

6.2 What each counter really queries

Counter	Query	Shows 0 when
Scheduled inspections today	bookableresourcebooking where _resource_value = user's resource, status = Scheduled (GUID), starttime within today	Resource not linked, wrong status GUID, or booking not for today

Counter	Query	Shows 0 when
Completed inspections today	msdyn_workorder where _duc_assignedresource_value = user's resource and duc_completiondate within today	duc_assignedresource or duc_completiondate not populated
Pending work order actions (offline version)	duc_inspectionaction owned by the user, status not 100000003/100000005, then work orders (statecode = 0) whose duc_primaryinspectionaction is one of them	Actions not owned by the user, or the work order does not reference the action

7. Offline specifics (mobile)

7.1 Mobile Offline Profile

Every table listed in section 2 must be added to the Mobile Offline Profile, with a filter that includes the records the inspector needs (own records + related organizational unit data). A table missing from the profile produces an empty dropdown or a silent failure with no error message.

Frequently forgotten: duc_organizationunitaccounttypes, duc_accounttype, duc_msdyn_incidenttype_msdyn_organizational, msdyn_incidenttypeservicetask, msdyn_inspectiondefinition, duc_processdefinition (and its stage/status/sub-status tables), duc_inspectionquestionconfiguration, duc_penalty, bookingstatus.

7.2 Query limitations the offline engine imposes

- \$expand is not supported — every lookup name is read with a separate request.
- The 'ne' (not equal) operator is ignored silently and returns all records — the control filters those cases in JavaScript.
- FetchXML link-entity / joins are not supported — queries are split into several calls (for example licence search, N:N incident types).
- Filtering on lookup columns is unreliable — results are filtered again in JavaScript.
- Records created offline are marked duc_iscreatedoffline = true so the server plug-ins return early after sync and do not duplicate the children the device already created.

7.3 Local cache (browser / device storage)

The control caches data in localStorage. After changing configuration, the user may keep seeing the old behaviour until the cache expires or the app data is cleared.

Key	Content	Expiry
MOCI_OrgUnit_Cache_{userId}	Organizational unit, unknown/site account, default incident type	1 minute
MOCI_User_ResourceID_{userId}	The user's bookable resource id	Never — must be cleared manually if the resource changes
MOCI_userCounts	Last known counters (used when offline)	Overwritten on each load
Inspection types / campaigns / incident types / vehicle types	Wizard dropdown values per organizational unit	Session-based; cleared with app data

First action for any "it still does not work after I fixed the data" report: clear the app data / browser site data for the user and reload.

8. Troubleshooting matrix

Symptom → most probable cause → record and field to check. Ordered by how often it occurs.

8.1 Home page

Symptom	Probable cause	Check / fix
Welcome banner shows "Inspector" or the English name	Arabic name not filled	systemuser.duc_usernamearabic
All counters show 0 or blank	User not linked to a bookable resource, or the cached resource id is stale	systemuser.duc_bookableresourceid; clear MOCI_User_ResourceID_{userId}
"Scheduled inspections today" always 0	Booking status GUID does not exist in this environment, or bookings are not in Scheduled status / not today	Booking status f16d80d1-...; bookableresourcebooking.starttime
"Completed inspections today" always 0	Completion date or assigned resource not set on the work orders	msdyn_workorder.duc_completiondate, duc_assignedresource
"Pending work order actions" always 0	Inspection actions not owned by the user, excluded status, or the work order does not reference the action	duc_inspectionaction owner and duc_status; msdyn_workorder.duc_primaryinspectionaction, statecode = Active
A list button opens an error or an empty list	The view GUID does not exist in this environment	Section 3.2 — re-point the view GUIDs
Start / End Patrol button not shown	No campaign matching all conditions	Section 6.1
Search returns nothing for an existing facility	The search reads the first 100 accounts and matches names that start with the typed text	Use Advanced Search (licence, PIN, building number) or type the beginning of the name
Unit-specific buttons (Natural Reserve / Wildlife) missing	Organizational unit English name does not match the expected text exactly	msdyn_organizationalunit.duc_englishname — see note in section 4

8.2 Multi-Type Inspection wizard

Symptom	Probable cause	Check / fix
Inspection type dropdown is empty	No junction rows for the user's unit	duc_organizationunitaccounttypes for duc_organizationunit = the user's unit
A type is listed but behaves wrongly (no identifier field, no search)	The numeric code is missing, so the control uses -1	duc_accounttype.duc_accounttype must contain the code of section 4.2

Symptom	Probable cause	Check / fix
Incident type dropdown is empty	No incident type linked to the unit, or all of them require judicial authority	msdyn_incidenttype.duc_organizationalunitid or the N:N table; systemuser.duc_hasjudicialauth
Campaign field is not displayed	The user's department hides it	msdyn_organizationalunit.duc_hidecampaignnonhomepage on systemuser.duc_department
Campaign dropdown is empty	Campaign not active / wrong type / wrong unit	new_inspectioncampaign: duc_campaignstatus = 2, duc_campaigntype = 100000000, statecode = Active, correct duc_organizationalunitid
"Please contact your administrator" on an Anonymous inspection (type 4)	Unknown account not configured	msdyn_organizationalunit.duc_unknownaccount
"Please contact your administrator" on a Site inspection (type 12)	Site account not configured	msdyn_organizationalunit.duc_siteaccount
"Please contact your administrator" when starting any inspection	No incident type could be resolved, or no department could be resolved	Select an incident type, or set new_inspectioncampaign.duc_incidenttype, or msdyn_organizationalunit.duc_defaultincidenttype; department comes from the incident type, then from the user
Qatari ID rejected	The control requires exactly 11 digits	Enter the full ID
A duplicate account is created for the same subject	The identifier field or the account type of the existing record does not match the search	account.duc_accountidentifier / duc_cpnumber / duc_moinumber and duc_newaccounttype
No address / GPS on the inspection	Location permission denied, or the request timed out after 30 seconds	Allow location on the device; the work order is still created

8.3 After the Work Order is created

Symptom	Probable cause	Check / fix
Work Order created but the form does not open	Form GUID does not exist in this environment	Section 3.2 — re-point the form GUID
Work Order has no Work Order Type	Incident type has no default type	msdyn_incidenttype.msdyn_defaultworkordertype
Work Order has no sub-status	Process definition incomplete, or the sub-status value is not a valid work order sub-status GUID here	duc_processdefinition.duc_defaultsubstatus → duc_processsubstatus.duc_value

Symptom	Probable cause	Check / fix
No process extension created (offline)	Incident type not linked to a process definition, or one of its default lookups is empty	msdyn_incidenttype.duc_processdefinition; section 5.1
No service task / no inspection form (offline)	No service task template, or no published inspection version	msdyn_incidenttypeservicetask rows; msdyn_inspectiondefinition with msdyn_state = 1
No questions on the service task (offline)	No question configuration for this incident type, or questions without a category	duc_inspectionquestionconfiguration.duc_primaryincidenttype and duc_questioncategory
No penalties on the service task (offline)	No penalty for this incident type, or tax type mismatch	duc_penalty.duc_incidenttype and duc_taxtype vs msdyn_workorder.duc_taxtype
No booking created automatically	The user has no bookable resource of type User, or no booking status named "Scheduled"	bookableresource (userid, resourcetype = 3); bookingstatus name = "Scheduled"
Duplicated children after sync (offline)	duc_iscreatedoffline not respected by the server plug-ins	Confirm the plug-in early-exit condition is deployed in this environment

9. Quick verification checks

Sign in as the user, open the browser developer console (F12) on the model-driven app and run the checks below. Replace {userId} with the user GUID (without braces). Each check should return at least one record.

User setup

```
/api/data/v9.2/systemusers({userId})?$select=duc_usernamearabic,duc_hasjudicialauth,_duc_organizationalunitid_value,_duc_department_value,_duc_bookableresourceid_value
```

Organizational unit setup

```
/api/data/v9.2/msdyn_organizationalunits({orgUnitId})?$select=duc_englishname,_duc_unknownaccount_value,_duc_siteaccount_value,_duc_defaultincidenttype_value,duc_hidecampaignhomepage
```

Inspection types of the unit

```
/api/data/v9.2/duc_organizationunitaccounttypeses?$filter=duc_organizationunit eq '{orgUnitId}'&$select=duc_name,duc_namear,_duc_accounttype_value
```

Incident types of the unit

```
/api/data/v9.2/msdyn_incidenttypes?$filter=duc_organizationalunitid eq '{orgUnitId}'&$select=msdyn_name,_msdyn_defaultworkordertype_value,_duc_processdefinition_value,duc_judicialauthorityisrequired
```

Bookable resource and booking status

```
/api/data/v9.2/bookableresources?$filter=_userid_value eq {userId} and resourcetype eq 3&$select=name  
/api/data/v9.2/bookingstatuses?$filter=name eq 'Scheduled'&$select=bookingstatusid
```

Patrol campaign of the day

```
/api/data/v9.2/new_inspectioncampaigns?$filter=owninguser eq '{userId}' and duc_campaigninternaltype eq 100000004&$select=new_name,duc_campaignstatus,duc_fromdate,duc_todate
```

Console diagnostics: the control writes detailed messages to the browser / device console with the prefixes [ProcessExtension], [WOSTHelpers], [WOI] and [SubStatus]. When a user reports a failure, ask for the console log — it names the exact record that could not be created.